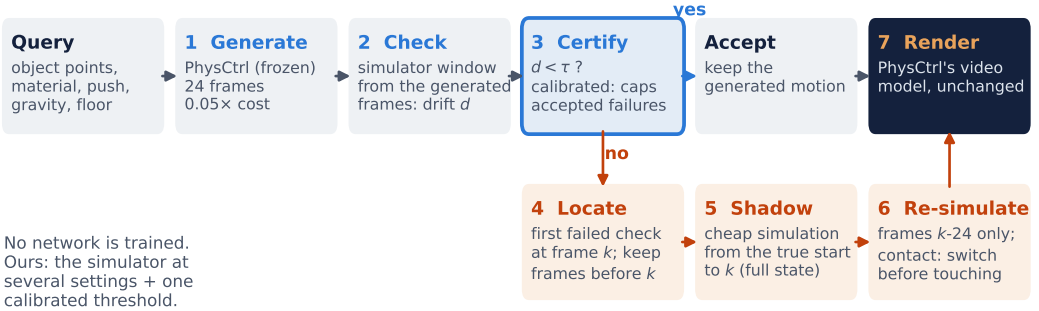
Query
object points, material, push, gravity, floor
1 Generate
PhysCtrl (frozen) 24 frames 0.05× cost
2 Check
simulator window from the generated frames: drift d
3 Certify
d < τ ?
calibrated: caps accepted failures
yes
Accept
keep the generated motion
7 Render
PhysCtrl's video model, unchanged
no
4 Locate
first failed check at frame k; keep frames before k
5 Shadow
cheap simulation from the true start to k (full state)
6 Re-simulate
frames k-24 only; contact: switch before touching
No network is trained.
Ours: the simulator at several settings + one calibrated threshold.
Steps 4-6 run only for rejected trajectories.